

Software - Interview Prep Document

At Cowlar, we believe our work should be a blend of hard work, innovation & excellence. This requires us to constantly strive to make ourselves better at what we do. This also makes it necessary for our hiring philosophy and process to be unique from what you're probably expecting.

Please read this entire document carefully. Please use the materials below to prepare for the interview.

Once you have completed your preparation and feel like you are ready, please email us with three preferred dates (s) of availability. We try to schedule the interview on your preferred dates¹. The venue for the interview/ test is our office located ([here](#)) in Islamabad. Our requirements, interview process & hiring philosophy are unique. It's important that you review the information carefully and decide if you want to work here.

General Requirements

- Hardworking, decent & honest
- Intelligent
- Should have the ability to assimilate complex ideas
- MUST be able to document your work and develop clear and easy to understand documents
- MUST understand how to create [loom](#) videos
- Recently graduating candidates MUST have the ability to **learn by doing** [\[1\]](#) [\[2\]](#) using online learning methods/tutorials
- Experienced candidates MUST have excellent communication skills & understand how to train, manage & help less experienced colleagues grow
- Must have the ability to follow instructions²
- Must be a team player [\[1\]](#) [\[2\]](#)
- Understand good work ethic [\[1\]](#)
- Must be a good fit into our team culture
- Must value your time & our time [\[1\]](#)
- Should treat everyone ³ with respect

¹ Please provide three (03) dates that work for you in order of preference. We try to schedule on your first choice date unless it's already booked.

² Please follow instructions provided to you during the interview, test and training phase. We don't like asking twice. This is especially important for fresh graduates.

³ Everyone means everyone, including our support & cleaning staff

- Must know how to use the internet ⁴
- Must have clear core concepts
- Excellent Programming & algorithm development skills
- Must have the ability to learn independently (without being asked)
- Must be good at reading documentation
- Experienced candidates must be good at writing documentation
- Must be able to summarize documents/tutorials/videos & teach core concepts to other team members
- Must be passionate
- Must enjoy work
- Must be ambitious & motivated

Hiring Process / Philosophy

Instead of going over multiple rounds of interviews (analytical, technical, HR, etc) with a candidate over a period of time, we believe it's best to do one long test /interview where the candidate spends time in our office answering questions & solving problems. You're welcome to ask us any questions ahead of time. You can also ask as many questions as you want **after** the interview.

Based on how the interview goes we may make an offer right away or get back to you in a couple of days. Alternatively, some candidates are placed on a wait-list and called when there is a requirement generated from our team.

For fresh graduates, we have some initial training requirements that need to be met before the selected candidate can begin work at our startup. The orientation part is usually a couple of weeks. You will typically spend 75% of your time going through the training program (learning the important things that will allow you to start contributing towards the actual work we do). The remaining 25% time will be spent on implementing what you learned (by doing small tasks/projects). As time passes you'll spend more time doing and less time learning.

This means that learning is an essential part of everyone's job. Not learning is *not an option here*. Everyone at our company typically spends about 25% of their time learning new things and working on cutting edge (sometimes bleeding edge tech). Our employees are expected to always be comfortable with learning new things, solving complex problems, and working as part of a team so we can deliver true value to our customers.

⁴ We don't expect you to know the answer to every question but we do expect you to know where to look. You should know how to google solutions to problems and use stackoverflow / online user forums / online community resources / discussion groups to your advantage. We expect you to know how and where to search for solutions to problems that you don't understand. We expect that before you ask a colleague, you will have tried your best to search for a solution online (by yourself)

We typically have a probation period of three to six months. We offer competitive salary packages and offer opportunities for upward mobility within the organization. Our salary packages are based on employee skills, learning ability, contribution to successful projects and how well you fit into the team culture.

Reasons to work here

- Opportunity to Learn from an exceptionally talented core team
- Our management style, processes, policies & team skills are constantly in the process of improving. Joining us will help you become part of a great team
- Access to purpose-built in-house specialized training modules to constantly make you better
- Access to paid online courses to accelerate growth & inspire you to build great products
- Freedom to work with core-team to choose an area of specialization over time that you excel in & that is mutually beneficial to you & the team. We want our people to love at least 90% of their work & the tasks they do.
- Opportunity to work in a conducive and comfortable work environment (We take care of our people, treat them with respect & expect the same in return)
- Free Meals & Snacks (Breakfast, Lunch, Dinner & unlimited Snacks). We have a Full-time dedicated Chef with a Full Kitchen in our office that cooks and serves fresh, clean meals for the entire team. Free Tea, Coffee, Snacks are available completely free of cost.
- Multiple opportunities for rapid professional & personal growth
- A Competitive Salary Package
- Lots of opportunities of upward mobility within our startup by demonstrating you can execute
- Opportunities to get multiple salary increments per year based on clearly defined goals / performance / contributions to team's core goals.
- The founders of Cowlar strive to be good leaders & lead by example [\[1\]](#)

How to Prepare for the Test

1. Prepare from the materials in this document. There are many links to useful resources. It is not expected that you will know everything on day one ⁵. We're just looking for people who can show the willingness to invest some time improving their skills before they come in for an interview. This is a win-win scenario for you (the candidate). Preparing for the interview/test can seem challenging at first. But if you dedicate some time and effort into learning and executing then it dramatically improves your chances of being hired asap. In the unlikely event you do not get an offer from us, you would still have learned some skills which can help you in future interviews and set you on a path of rapid career development. Our test is **not easy**. This is because we are looking for the best people. We're not looking for button pushers, clock watchers, or people who come to work as a compulsion / necessary
- 2.

⁵ Please note. If you're an experienced developer, the threshold of what you should know will be different. We don't care about the number of years you've worked in the Pakistani software services industry, we want to see what happens when the rubber hits the road. Everyone says they're good. Please show us, don't tell us.

3. chore. We are looking for talented and passionate people who love their field of work and enjoy solving difficult and challenging problems.
4. **It is perfectly fine to reschedule your interview if you think you need more time to prepare.** There is no negative marking for delaying or rescheduling the interview date. You can reschedule as many times as you want ⁶. That's perfectly ok from our point of view. It is NOT OK to show up unprepared and waste your & our time. People who show up unprepared are quickly offered a cup of tea & their interview ends in no time. We only spend time interviewing people if we think there is a chance we can hire them. We generally only interview candidates ONCE.
5. You can use the internet during the interview. There is no negative marking for using the internet during the interview. You can also access this document (or any bookmarks, code snippets, previous projects during your test for reference materials. You can bring as many cheat sheets with you as you want.
6. It is generally recommended that you bring your own laptop so it's installed with your preferred software/ code editors etc. If you cannot bring your own device, one will be provided to you.
7. As you watch the videos in the test materials section below, increase the playback speed to **1.25x**. **Then as you get comfortable increase speed to 1.5x and eventually to 2x speed.** Eventually, you'll be able to watch a 40 minute lecture in 20 minutes without a problem.
8. Look at the [1] level links for each section before diving in. Sometimes this will not be the case, you need to realise when you need to abort / close a link and move on to the next link.
9. During the interview and demo portion at our office. You can use two life lines.

(A) **Can use the internet.** If you're experienced, you should have a decent hold of syntax (you should not have to look at a website in order to write a for loop)

(B) **Phone a Friend/Person** (You are allowed to call anyone you want (you will have 5 minutes to have a conversation over the phone)
10. Please use online resources to revise and practice your core concepts. Here is a little secret. If you follow it, we will hire you. Worse case scenario is you become better skilled at what you do. The secret is to work hard and spend the necessary time learning. We are telling you in advance what questions we will ask and where/ how to prepare for them.
11. You need to get comfortable with learning online and offline. One of the benefits of working at Cowlar is having paid resources and training materials available to you. If we hire you we will have purchased more than 100+ paid courses / online tutorials from sites like Udemy, VueMastery, VueSchool, SkillShare, Pluralsight etc. so our team has the best conducive environment for learning. We also arrange to provide webinars that might be useful for your professional growth.

⁶ If you want to reschedule your interview on the same day it was supposed to take place, you'll need to give us four hour's advance notice. This helps us differentiate from people who want to reschedule & those who don't show up.

Test Preparation Materials

Please note. As we receive hundreds of applications for each position, ultimately it might come down to who can answer/demo the most skills (practically) from the questions/topics below.

Please do not ask to be interviewed if you have not prepared from the materials below

TABLE OF CONTENTS	
No	Topic
1	<u>HTML / CSS / Bootstrap / Tailwind</u>
2	<u>JavaScript</u>
3	<u>NodeJS</u>
4	<u>VueJS</u>
5	<u>Git</u>
6	<u>Misc</u>
7	<u>Flutter</u> (Bonus)

JavaScript

Take [this](#) JS crash course to revise.

Your interview questions will be from the following links

[\[0\]](#) [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#) [\[6\]](#)

1. What is the Event Loop [\[1\]](#)
2. How would you make a copy of an array [\[1\]](#)
3. JSON, XML & AJAX [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)
4. DOM [\[1\]](#) [\[2\]](#)
5. Push, Pop, Shift, Unshift
6. Closure in JS [\[1\]](#) [\[2\]](#)
7. Array manipulation [\[1\]](#) [\[2\]](#) [\[3\]](#)
8. Higher order functions on arrays (map for JS) [\[1\]](#) [\[2\]](#)
9. Higher order functions - Filter & reduce method [\[1\]](#) [\[2\]](#) [\[3\]](#)
10. Functional programming [\[1\]](#)
11. String Manipulation
12. Parsing strings [\[1\]](#) [\[2\]](#) [\[3\]](#)
13. Loading JSON data from a URL (Async Callbacks) [\[1\]](#)
14. Axios [\[1\]](#) [\[2\]](#)
15. Sockets [\[1\]](#)
16. Can I use Javascript to detect if there is an internet connection?
17. When a user is logged in, How would you save their authentication information?
18. What can sockets be used for?
19. How to get the value of a certain key from a Javascript object? [\[1\]](#)
20. Decoding encoded bit data from a variable
21. How would you remove duplicate values from an array [\[1\]](#) [\[2\]](#)
22. What is the **'this'** keyword in JavaScript?
23. What are exceptions? [\[1\]](#)
24. How are exceptions handled in JavaScript? [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)
25. Which is faster if/else statement or switch case (and why ?)
26. Must know how to debug JavaScript [\[1\]](#)
27. LinkedList [\[1\]](#)
28. Difference between Callbacks vs Promises vs Async/await? [\[1\]](#) [\[2\]](#)
29. What are the drawbacks of Callbacks? [\[1\]](#) [\[2\]](#)
30. Promises
31. What is a drawback of Promises
32. Cache

- 33. AMP [\[1\]](#)
- 34. Offline Canvas in Chrome [\[1\]](#)
- 35. Service workers & Caching [\[1\]](#) [\[2\]](#) [\[3\]](#)
- 36. The Service Worker Lifecycle [\[1\]](#)
- 37. Debugging Service workers using DevTools [\[1\]](#)
- 38. How can we compress data in Javascript [\[1\]](#)
- 39. Create an interactive JavaScript 3d Model [\[1\]](#)
- 40. Build a 360-view image slider with JavaScript [\[1\]](#)
- 41. What is the difference between a linter and a formatter?
- 42. What formatting tools are available [\[1\]](#)

NodeJS

Please look at this [\[Link\]](#)

- 43. What is Node [\[1\]](#)
 - 44. What is NPM ? [\[1\]](#) [\[2\]](#) [\[3\]](#)
 - 45. Callbacks & Promises in NodeJS [\[1\]](#) [\[2\]](#) [\[3\]](#)
 - 46. Faster Async functions and promises [\[1\]](#)
 - 47. Sequelize.JS (ORM) [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#) [\[6\]](#)
 - 48. TypeORM [\[1\]](#)
 - 49. What are migrations [\[1\]](#)
 - 50. What are seeds [\[1\]](#) [\[2\]](#)
 - 51. Building a RESTFUL API [\[1\]](#) [\[2\]](#)
 - 52. Auto Generating API docs with Node & Swagger [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)
 - 53. Garbage collection [\[1\]](#)
 - 54. How can we use migrations [\[1\]](#)
 - 55. Express.js Framework For Node.js [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)
 - 56. Fastify [\[1\]](#)
 - 57. What is a framework [\[1\]](#)
 - 58. Fastify vs Express [\[1\]](#)
 - 59. What is NestJS [\[1\]](#)
 - 60. What are Race Conditions and how to handle it [\[1\]](#)
 - 61. Push Notifications using NodeJS + web-push [\[1\]](#)
 - 62. Unit testing (Jest, Mocha, Jasmine)
 - 63. Bundlers, Task runners, Build / Dev tooling [\[Grunt, Gulp, NPM, Webpack\]](#)
 - 64. Google Sheets for Node JS developers [\[1\]](#)
- [\[50 Interview questions & answers for NodeJS\]](#)

VueJS

Don't know VueJs? Click [here](#)

[\[Program with Erik\]](#)

[\[Why do FrontEnd with VueJS\]](#)

[\[Vue over react\]](#)

[\[Vue.js 2 in 60 minutes\]](#)

[\[Learn Vue.js in three hours\]](#)

[\[10 Tips to become a better VueJS developer\]](#)

[\[Intro to Vue - Build a to-do app in 20 minutes\]](#)

- 65. Why Choose VueJS or React? [\[1\]](#) [\[2\]](#) [\[3\]](#)

66. What are VueJS components
67. How are VueJS components created
68. VueJS Components LifeCycle
69. Routing in VueJS
70. What is the difference between **v-if** and **v-show**
71. What is the **updated** function (what are its pros and cons)
72. Filters in VueJS
73. Fragments in Vue [\[1\]](#)
74. Plugins
75. Mixins [\[1\]](#)
76. What is single Page Applications in Vue
77. Declarative rendering
78. Class and style binding
79. Event handling
80. How to pass data between components
81. Props
82. Type of loops in VueJS
83. How many ways can props be declared
84. What are the pros and cons of using those different ways
85. Attributes
86. How are attributes used
87. Vue Apollo (GraphQL with Vue) [\[1\]](#)
88. Vuex [\[1\]](#) [\[2\]](#)
89. Vue-DevTools [\[1\]](#)
90. Debugging VueJS [\[1\]](#)
91. VueJS Dropdown with Keyboard Accessibility [\[1\]](#)
92. VueJS search with vue-fuse [\[1\]](#)
93. How are HTTP requests sent via Vue
94. Vue Testing with Jest [\[1\]](#)
95. Unit testing [\[1\]](#)
96. State management in Vue
97. Lifecycle Hooks
98. Static site generation for Vue (Gridsome) [\[1\]](#) [\[2\]](#) [\[3\]](#)
99. Using Gridsome with Wordpress [\[1\]](#)
100. Server-side rendering (SSR) with VueJs
101. Faster Web applications with Vue 3 [\[1\]](#) [\[2\]](#)
102. Vue 3 Composition API [\[1\]](#)
103. Vite

HTML5 /CSS /BootStrap /Jquery

[Beginners HTML5]

104. Demonstrate excellent proficiency in HTML5, CSS3 [\[1\]](#) [\[2\]](#) [\[3\]](#)
105. What is HTML canvas [\[1\]](#) [\[2\]](#) [\[3\]](#)
106. Speed up HTML with Emmet [\[1\]](#) [\[2\]](#)
107. Bootstrap 4 [\[1\]](#)
108. What is an offset in Bootstrap [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)
109. Media queries in Bootstrap [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)
110. What is z-index (CSS) [\[1\]](#) [\[2\]](#)
111. Standard media queries [\[1\]](#)
112. CSS transitions [\[1\]](#)
113. Drawing + Animations on Canvas
114. What is Webkit in CSS [\[1\]](#) [\[2\]](#) [\[3\]](#)
115. CSS Grid [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#)
116. Flexbox (CSS) [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)
117. Responsive design (viewport & media queries) [\[1\]](#) [\[2\]](#) [\[3\]](#)
118. CSS3 Viewport [\[1\]](#) [\[2\]](#) [\[3\]](#)
119. I have 10 rows in a table, how can I change the color of the 4th row without using a class
120. Jquery [\[1\]](#) [\[2\]](#) [\[3\]](#)
121. Why do we get \$undefined errors [\[1\]](#)
122. How can you play a video using HTML5 [\[1\]](#)
123. What are iframes [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)
124. How can you add 360 degree images on HTML page [\[1\]](#)
125. Tricky CSS properties (Box Model,Font-relative, viewport etc) [\[1\]](#)
126. Must understand and know CSS tricks [\[1\]](#) [\[2\]](#)
127. Knowledge of CSS Preprocessors (Ideally Sass) [\[1\]](#) [\[2\]](#)
128. What are CSS image sprites [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#) [\[6\]](#)
129. 3d models in HTML documents [\[1\]](#) [\[2\]](#)
130. Can I build a 360 view of an object using HTML
131. Basic understanding of Google Developer Tools [\[1\]](#)
132. How to improve load performance using Chrome DevTools [\[1\]](#)
133. Supercharging Page load [\[2\]](#)
134. Designing for speed & hacking user perception [\[1\]](#)
135. Advanced Chrome page load speed [\[1\]](#)
136. Charting Libraries D3, HighChart [\[1\]](#)

Git

- 137. Git & Github Crash Course [\[1\]](#)
- 138. Github [\[1\]](#)
- 139. What is GitFlow [\[1\]](#)
- 140. Resolving conflicts during Git merge [\[1\]](#)
- 141. Git Graph [\[1\]](#)
- 142. Git rebase [\[1\]](#)[\[2\]](#)
- 143. Git Submodule
- 144. Git squash

Database

- 145. MySql
- 146. PostGres
- 147. mongoDB
- 148. Sqlite
- 149. Indexing

Misc Section

- 150. Linux basic command line [\[1\]](#) [\[2\]](#) [\[3\]](#)
- 151. Bash [\[1\]](#)
- 152. SSH [\[1\]](#) [\[2\]](#)
- 153. VIM [\[1\]](#)
- 154. HTTP Crash course [\[1\]](#)
- 155. Google Chrome Dev tools crash course [\[1\]](#)
- 156. **What is a REST API?** [\[1\]](#)
- 157. **REST vs GraphQL** [\[1\]](#)
- 158. REST vs RPC vs GraphQL [\[1\]](#)
- 159. **GraphQL** [\[1\]](#)
- 160. **MQTT** Protocol [\[0\]](#) [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#)
- 161. **MQTT** Mosquitto broker [\[1\]](#)
- 162. **MQTT** Publishers / Subscribers [\[1\]](#)
- 163. tRPC [\[1\]](#)
- 164. Sending Curl
- 165. Nginx [\[1\]](#)
- 166. WebSockets [\[1\]](#)
- 167. What are Progressive Web Apps (PWAs)?
- 168. What is Webpack? [\[1\]](#) [\[2\]](#)
- 169. What's the difference between Webpack / Gulp / Grunt ?
- 170. What is tree shaking?
- 171. Must understand MVPs [\[1\]](#)
- 172. What is docker [\[1\]](#)
- 173. Must understand the difference between prototyping and building for production
- 174. Basic understanding of DevOps
- 175. Basic understanding of Kubernetes / Spinnaker / Pipelines
- 176. Will prefer someone with an understanding of software testing
- 177. Working knowledge of Git (Github / bitbucket etc)
- 178. Working knowledge of using Google Docs
- 179. Working knowledge of using Slack
- 180. Be able to create short videos/screengrabs using Nimbus Screen recorder or any alternative software. Must be able to quickly (and asynchronously communicate with the team via Slack / Whatsapp / short videos / well-written code & Good documentation.
- 181. Must be able to demonstrate ability to work as a team player. This means you should have good coding practices and should document/comment on your code properly. Being a good team member also means the ability to communicate effectively.

Here are some free resources that can help you. For fresh candidates, it's okay if you don't know everything. For experienced candidates, we expect more.

[\[Net Ninja\]](#)

[\[Dylan Israel\]](#)

[\[Coding Garden with CJ\]](#)

[\[Coding train\]](#)

[\[Functional JS with Fun Fun Function\]](#)

Flutter (Optional)

Here is a free Flutter [playlist](#) that is very good to learn and revise from. If we hire you we have multiple paid courses that you'll be able to access.

[\[How Flutter works in 8 min\]](#)

[\[Flutter - Widget Tree & Element Tree in 8 min\]](#)

[\[Flutter Cookbook\]](#)

182. Why Flutter, tell 5 pros and cons of flutter?
183. Why does Flutter use Dart ? [\[1\]](#)
184. Basic Dart programming and tools. [\[1\]](#) [\[2\]](#) [\[3\]](#)
185. Abstraction, Inheritance, polymorphism and encapsulation concepts in dart.
186. What is Pubspecc.yaml file
187. Difference between Stateful and Stateless Widgets?
188. Asynchronous programming in flutter?and why do we need that?
189. What is Await in flutter?
190. Difference between then() and whenComplete() function?
191. Use of Native widgets for IOS and Android.
192. Should be able where and how to add Dart packages in flutter app.
193. How to pass data/variables across widgets in flutter app.
194. Use of Futures,Async, Await and timeout in flutter.[\[1\]](#)
195. Difference Between async and async* ?
196. State Management in flutter via Getx, Provider and other methods.
197. What is Isolates ?
198. JSON Manipulation (parsing, encoding) in flutter.
199. Knowledge about Web APIs (REST, SOAP) and consuming API calls in flutter.
200. Knowledge about GET,POST, PUT,CURL requests.
201. Should have knowledge about SQLite (or SQflite plugin in flutter) and all the basic database concepts and operations(CRUD).
202. Material App and how to use it in flutter.
203. Understanding of Android Studio project build, xml and other files, where to add permissions and dependencies for android and IOS project.
204. Callbacks in flutter.
205. Deep Understanding about Lists and Maps in dart.
206. Basic knowledge about firebase or any other commonly used cloud service to integrate with flutter app for the data exchange between app and cloud, or for push notifications etc.
207. Secrets in Flutter  [When to Use Keys - Flutter Widgets 101 Ep. 4](#)
 [How To Store API Keys For Staging & Production Environments](#)
208. Navigation to different screens in flutter (routes concept). [\[1\]](#) [\[2\]](#)
209. Local Push Notifications [\[1\]](#) [\[2\]](#)

- 210. How can you communicate Between Dart and Native code in flutter?
- 211. What is Pump and PumpSettle in flutter Testing?
- 212. Flutter Dev tools [\[1\]](#)
- 213. What is flutter for Web [\[1\]](#)

MUST PREPARE FOR THE INTERVIEW

Must have clear concepts!

- 1. GIT
- 2. Unit Test and Integration testing
- 3. Docker
- 4. API Integration
- 5. .env variables
- 6. ORMS
 - a. Sequelize
 - b. Prisma
 - c. Mongoose
- 7. ADDITION POINTS**
 - a. MQTT
 - b. Influx
 - c. GitLab - CI/CD
 - d. Github Actions

DO NOT LEAVE THIS SECTION

e.

Demo

Demonstrate your skills (optional)

Any projects you have done, bring them with you and showcase your code / etc on your laptop. This can include your professional projects, FYP, your python work, github contributions. Anything that showcases your ability. We are particularly interested in seeing how cleaning / well the code is written (do you write comments in your code etc)

Interview Tips

Here are some resources to use to reduce interview nerves [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#). We highly recommend watching all these videos.

When to schedule an interview

This is a question we often get asked. This will depend on the candidate after they have had a chance to go over the test preparation materials. Generally speaking ...

For **fresh graduates**, it is highly recommended to invest some time in learning the material. Take a look at the topics, watch the videos / read the articles and **learn by doing**. You will remember what you learn *by doing*. When you feel like you have sufficient ideas and are 2x or 5x better than you were before you started your preparation, please contact us by email to set up a date. Please send us three dates that work for you. Do not think that there is a deadline for this position. Don't schedule the interview fearing that someone else will take this position. Schedule it when you're ready because we usually only interview candidates **ONCE**⁷. If you're good and promising, there is a very good chance we will hire you and put you through our training program. In the unlikely event where you don't get an immediate call, you might be placed on a short-list where we tap into first when looking to add new fresh graduates.

In the unlikely event that you don't hear back from us at all, (the worst case scenario), you will always be in a win-win position. This is because you will

- Still have gained some important & useful skills
- These skills will help you throughout your professional career
- Dramatically increase your chances of being hired by a good company in the future.
- You'll learn how to learn online. That is an important asset.

⁷ We do make exceptions to this rule in exceptional circumstances. The chances of you getting that exception are about 2%

For **experienced professionals**, please review the materials carefully. You might know a lot of these things. Our test / questions get increasingly difficult based on the duration of your industry experience and based on the reputation of the (good) companies you worked for in the past. You might learn something new while you revise and get better at what you do. There is really no reason to not learn.

From our point of view, the good thing about hiring experienced professionals is that they have already learned a lot and can begin contributing to our team in a short period of time. Sometimes, the bad thing about hiring experienced professionals is that they have learned some bad habits ⁸ in the past. Please leave all those bad habits behind when you come in for the interview. This can be difficult to do. Bad habits are difficult to break. But one can always try. It takes self-realization, will power & determination. The greatest minds are those that *can be* changed. We don't hire people with bad habits. The reason is that we cannot allow anyone to ruin our culture, no matter how brilliant they are. There is no & never will be any place for *brilliant jerks* at our startup.

⁸ The bad habits can be programming or development related. They can be related to poor development methodologies or practices. They can also be related to not being a team player. They can be related to not acting responsibly. They can be related to not treating others with respect. They can be related to gossiping or participating in office politics. If you think writing good maintainable code, commenting, documentation, Testing, Software Quality Assurance is for losers, please schedule an interview for Feb 30, 2050

Interview FAQs

1. What's the biggest mistake people make during the interview process?

They don't invest enough time in learning from the test materials or in preparation for the interview. Sometimes, they just ask to be interviewed too early. Showing up with not enough adequate preparation reduces your chances of being offered a position exponentially.

2. What time does the interview usually begin?

The typical time for starting an interview is at 11.30 am. You pick the date, we pick the time (unless you're traveling from outside the city. In this case, please inform us beforehand in writing so we can accommodate you)

3. How long does the interview/test last?

The duration can be anywhere between 5 minutes to 5 hours. Please plan on being here for 5 hours in case the interview goes that long.

4. What to expect on interview day?

- You arrive between 11.15 am to 11.25 am. You will be judged for being late. Please inform us ahead of time in case you're running late. Please plan to arrive on time and consider traffic / your commute to our office.
- Your first question is water/coffee/tea or green tea? Please pick one of the choices. Nothing is one of the choices. This is a clear sign that you read this document carefully and are capable of following instructions.
- You are then given a piece of paper with some analytical questions and a pen + calculator. Please answer the questions. You can answer any 5 questions of your choice. If you want to use the internet, show us the questions so we can keep a score of how many without internet and how many with the internet.
- You're then asked a few simple questions about yourself and regarding the things mentioned in the CV.
- You will be given 3 to 5 minutes to present your case (pitch us why we should hire you and how your preparation went)
- We will then start asking questions. The questions and how you chose to answer them will determine how many people from our team join/leave the interview.
- The questions tend to increase in difficulty level over time. We might ask you to write some code.
- Please keep your answers short, concise, and to the point. We do appreciate it when someone explains a related concept with great clarity, but please don't go around in circles for every answer.
- Please try to answer all questions honestly and don't hesitate if you don't know the answer. Take a couple of seconds to think or ask us to repeat the question if you had difficulty understanding the question.

- Please don't think there are any bonus points for not using the internet. You can actually use the internet every time you're asked a question. In this case, we are judging (& hoping) that your answer is **correct** and taking note of the duration of how long it took you to answer. It's better to be late and correct.
- If we end up spending more than one hour, you can ask us how your interview went at the end (we only provide feedback optionally / if you ask us). We are also looking for feedback about the interview process and how you think we can improve as a startup.
- At any time, you can choose to decide that you don't want to work here and leave.
- Please feel free to use the toilet (in case you need to)
- Please feel free to pause the interview in case you need to pray. You can pray in the designated area in the office.
- Lunch, snacks, and dinner are offered free of cost. Please accept food /drinks that are offered and don't do 'takulf'.
- If you smoke, no smoke breaks are allowed during the interview. You are welcome to go outside and smoke as much as you want before or after the interview.
- Please treat our people with respect. We try very hard to do the same.
- Please keep all comments/feedback to yourself till the end of the interview (in case the interview lasts more than one hour).

5. Why can the interview duration last 5 hours?

As we spend more time interviewing a candidate, the difficulty level of the questions increases. Please understand that we only interview/test people for longer because they are overcoming the obstacles and increasing their chances of getting an offer on the spot (or very soon).

6. Do you provide any remuneration for candidates traveling from outside Islamabad / Rawalpindi?

No

7. Why is this document so long?

We want to know if you can read, listen, understand & execute. We also want to know if you actually take the time out to read this entire document. We will be testing this during the interview. The length of this document is also meant to discourage people from asking to be interviewed in-person (esp who are not interested in working hard / reading documentation / learning new things). Your job as a software engineer requires you to know how to read well and how to write well too (although that is a much harder skill to master). Reading this document carefully is part of the test / interview :)

8. Will you answer my questions about the YC experience or provide feedback about my startup idea?

No

9. What are you looking for in fresh candidates?

In fresh candidates, we are looking for people who want to become polyglot software engineers /developers. During your initial training period, you will be trained in many facets of software development & software engineering. You must be okay with developing a decent understanding/working knowledge of various aspects of technology development (basically you need to know what other people are doing in the company so you can talk to them if you're in need of their expertise for your project/task).

10. How do you decide on Salary?

Our salary packages are based on a combination of employee skills, learning ability, ability to execute without mistakes, their contribution to successful team projects & how well you fit into the team culture. Simply put, the more you know and the faster you can learn is directly related to how fast you can execute. If we can ship good, reliable software faster, we can make more money. We can then share that (extra) money with you.

11. How do you decide which roles to assign to new team members once you hire them?

Generally speaking, over time, you (the employee) & us (the company) will mutually agree on your areas of specialization based on your preferences, skills, your proficiency, and the company's needs after the orientation and training period.

12. Best piece of advice to prepare for the interview/test?

Please try to relax. Try to avoid being anxious. We have made a special effort not to make the interview tense. This is because we want to see how you perform in a real job-like condition. Just prepare as much as you can, try your best and leave the results to Allah. Here are some resources to use to reduce interview nerves [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#). We highly recommend watching all these videos.

This is the training section. Ignore this for now.

Cowlar Software (Web Dev track) training [Document](#)

HOW TO APPLY FOR THE JOB

Before applying, just know that our process is highly competitive, but for those who have prepared well will sail through the process and won't even notice it, but those who are applying blindly will have a tough time.

We really appreciate people who value their time, as we value yours and ours time.

How to apply?

It is quite simple, just go the work.cowlar.com,
Find your relevant job under the, "Open Job," section and apply!

Follow the simple steps as guided in the application form and you should hear back from us, if you are shortlisted. Next steps will be guided by the Talent Acquisition team via email or call.

In case, you have an issue with the job application, or there is an error, or you have any queries, you can write to us at, careers@cowlar.com, but please make sure you add relevant subject, and craft your email properly, a well written email can get you much faster response.

Best of luck for your application.